

VENKATA GANJI

AI PLATFORM SRE | GPU MODEL SERVING & KUBERNETES
DISTRIBUTED SYSTEMS | INGESTION, RETRIEVAL & RELIABILITY



Concord, CA | 925-322-2650 | venkataganj.careers@gmail.com | No sponsorship required
linkedin.com/in/venkata-sai-ganji-3a30a2a4 | Architecture Portfolio <https://offroadguy.github.io/>

AI Platform SRE and Distributed Systems Engineer with 12+ years of production experience. Owned Kubernetes operations, vector databases, infrastructure automation, observability, and recovery capabilities within an internally engineered enterprise AI platform; co-owned GPU model-serving tooling and model-weight onboarding.

ENTERPRISE AI PLATFORM AND MY CONTRIBUTION

PLATFORM SCOPE Enterprise chat Document intelligence Agent applications Studio UIs & APIs		
Ingestion & documents Connectors, parsing, chunking, proprietary document conversion	Retrieval & agents Search, embeddings, memory, orchestration, MCP/tools, guardrails	GPU model serving Open-weight models, vLLM, weight staging, LoRA adapters
Redis MongoDB Solr Elasticsearch/OpenSearch Milvus Kafka Object storage		
Platform operations: Kubernetes, GitOps/IaC, metrics/logs, probes, alerts, CronJobs & recovery		
OWNED Kubernetes operations, vector DBs, IaC/ops tooling, observability, capacity/cost, selected-platform DR		
CO-OWNED GPU serving infrastructure and model-weight onboarding		
Platform footprint: AWS GCP Alibaba Cloud Apple private cloud		
250+ nodes / ~200 TB Vector database infrastructure	~7.0s to ~1.9s Vector-query latency	\$266K+ annualized Solr infrastructure savings

PROFESSIONAL EXPERIENCE

	Apple GenAI Platform Engineering & Cloud Architect	Apr 2024 - Aug 2026
Contract via Mphasis Corp Enterprise AI infrastructure and platform operations		
- Owned Kubernetes/EKS operations and deployment tooling for ingestion and inference: upgrades, routing, autoscaling, resource sizing, and production diagnosis. Productionized 10+ GenAI microservices across Dev/UAT/Prod. - Developed and co-owned GPU model-serving tooling using Helm/Argo CD and custom vLLM images; configured tensor parallelism, GPU memory budgets, context limits, and LoRA. Built S3-to-FSx weight staging with checksum validation and read-only serving mounts. - Owned vector database capacity and reliability across Solr, Milvus, and OpenSearch; scaled to 250+ nodes / ~200 TB and cut query latency from ~7.0s to ~1.9s through JVM/GC, OS, cache, query, and index tuning. - Authored and owned Solr IaC and lifecycle automation with Terraform, CloudFormation, Ansible, and Python; built tooling for health checks, backups, recovery, and stale-resource cleanup. - Owned observability from metrics and logs to dashboards and alerts; helped reduce incidents from ~20/day to ~2/week through automation, performance tuning, and recurring-failure remediation. - Migrated 200 Solr nodes to AWS Graviton/ARM64, preserving throughput while saving 19.8% (\$266K+ annualized); productionized EC2 Mac document conversion with application developers, saving ~\$16K/month.		

CLOUDWICK TECHNOLOGIES INC | APR 2014 - APR 2024

 **Amgen**
AWS Cloud Solutions & Migration Engineer

Sep 2023 - Apr 2024

Cloudwick client engagement

- Migrated **100+ AWS accounts** to Control Tower and Organizations using repeatable Python, CloudFormation, CDK, and SSM workflows; implemented secure S3 migrations with IAM/KMS.
- Tuned vector databases, resolved GenAI ingestion failures, and built Kubernetes deployment pipelines and Python monitoring integrations.

 **Freddie Mac**
AWS Cloud Solutions Architect & Infrastructure Consultant

Jan 2019 - Aug 2023

Cloudwick client | Technical leadership for 3-8 engineers across 5+ projects

- Led **three mission-critical migrations** across distributed data and live OLTP/web platforms, owning architecture, data movement, performance testing, and production transition.
- Modernized a mortgage-document platform from Hadoop/Solr to Elasticsearch/OpenSearch and S3; led Spark migration jobs for Solr-to-Elasticsearch and HBase-to-S3 data movement.
- Designed ingestion frameworks with Databricks, AWS Glue, Talend, and S3; built reusable Terraform, CloudFormation, Ansible, and Python/Airflow provisioning and lifecycle automation.
- Delivered hybrid-cloud disaster recovery within six months with **~15-minute RPO** and **~2-hour RTO**, using cross-region replication, S3, Airflow, Python, and Elasticsearch.

 **Cyber Data Lake / Amorphic**
Lead Solutions Engineer

Nov 2016 - Dec 2018

Cloudwick | Cybersecurity telemetry and distributed analytics

- Built telemetry ingestion correlating PCAP, NetFlow, application logs, and firewall events across Hadoop, Kafka, Spark, Solr, HBase, and Elasticsearch.
- Designed a three-rack IDS/IPS and packet-broker test bed; deployed and tuned 40 Gb/s packet-capture technology and built Prometheus monitoring for collector agents.

EARLIER DISTRIBUTED SYSTEMS DELIVERY**Selected clients | Senior Operations Solutions Engineer | Feb-Oct 2016**

Daimler, SimplyMeasured, Classmates / The Battle Group: Hadoop builds, Ansible automation, security, benchmarking, and production readiness.

Amobee / Turn | Operations Solutions Engineer | Apr 2015-Jan 2016

Operated Hadoop/HBase at up to ~400 nodes per cluster and 30 PB total capacity; delivered HA/DR, fleet refreshes, and Python/JMX monitoring.

Limelight Networks | Operations Engineer | Apr 2014-Apr 2015

Designed HA/DR for Hadoop billing, maintained ~80-node video analytics, and tuned Hive/MapReduce.

TECHNICAL EXPERTISE**AI & orchestration:** vLLM, LoRA, Helm, Argo CD, Kubernetes/EKS, Kustomize, Docker, S3, FSx for Lustre**Data & performance:** Solr, Milvus, OpenSearch/Elasticsearch, Hadoop/HDFS, HBase, Kafka, Spark; JVM/GC tuning**Automation & reliability:** Python, Bash, Terraform, CloudFormation, Ansible, AWS, Prometheus, Grafana, Splunk; HA/DR**EDUCATION**

Bachelor of Technology, 2011.

Graduate coursework in Computer Science: Texas A&M University, 15 credits (2014); Montana State University, 15 credits (2013).